

LOGAN SMITH - TECHNICAL DESIGNER

Orlando, FL | (305) 984-6021 | logansmithnryan@gmail.com | [Portfolio](#)

SECURITY CLEARANCE: ACTIVE SECRET Since Jan 2025

TECHNICAL SKILLS

- **Game Engine & Tools:** Unreal Engine 5 (Blueprints, C++, Editor Utility Widgets, Animation Notifies, Behavior Trees, State Trees), Perforce, Git, Bitbucket.
- **Programming & Scripting:** C++, Python, Java, DAX, JQL.
- **QA & Telemetry:** Test Automation (Selenium, Behave, Cucumber/Gherkin), Performance Profiling, Jira, Confluence, Power BI.
- **Methodologies:** Agile, Scrum, CI/CD Pipeline Integration, Systems Architecture.

PROJECTS

Combat Designer | UNSANCTIFIED (Professional) | Contracted | Unreal Engine 5 | Team Size: 23

- Designed and implemented the complete weapon system and weapon pipeline, including all in-game weapons and attack logic.
- Set up animation-based combat attacks using Animation Notifies for precise hit registration and attack-cancel windows.
- Designed and implemented all player character movement, locomotion blending, and animation state machines.
- Iterated on the 3 C's (Character, Controls, Camera) across multiple milestones.
- Refactored the codebase for scalability across the team, improving navigability and reducing integration friction.
- Collaborated with the art team on animation timing, combat feel, and locomotion transitions through continuous cross-discipline iteration.
- Designed enemies scoped to fit and challenge within the established combat sandbox.

Technical Designer & Team Lead | OFF AIR (Published) | Duration: 4 Months | Unreal Engine 5 | Team Size: 5

- Built a reaction-based projectile parry system using Blueprint Interfaces and polymorphism, eliminating expensive casting chains at scale.
- Designed and implemented all 5 enemy types and a boss, each with distinct behavioral states, attack patterns, and AI logic.
- Designed and implemented the full player movement suite: Wall Jump, Wall Cling, Slide, Dash, and Ground Pound.
- Engineered a health orb object pool (capped at 25) to stabilize actor count during high-density combat encounters.
- Built a dedicated weapon-swap tracking actor to handle stowed weapon ammo state without requiring persistent weapon actors in the world.
- Created an Editor Utility Widget (EWBP) enabling designers to configure enemy waves, spawn points, and per-enemy health values without editing Blueprint arrays.
- Led a team of 5 through Agile sprints; owned milestone planning, task delegation, and project management.

Combat Designer | Beast of the West (In Progress) | Unreal Engine 5 | Team Size: 3

- Designed and implemented a *Ghost of Tsushima*-inspired stealth and AI perception system with tiered vision zones and suspicion thresholds.

- Blended a custom C++ State Machine with UE5 Behavior Trees and AI Perception for performant, flexible enemy AI logic.
- Engineered forgiving vision margins (2x detection at center, 0.5x at periphery) to create readable, learnable stealth gameplay.
- Added vertical blind spots to prevent AI from detecting players directly above or below, enabling rooftop assassination and vertical gameplay.
- Built a Parry/Poise System and dedicated combat camera for 1v1 engagements, most prominent in the boss battle.
- Utilized CommonUI for all UI/UX systems and built scalable traversal systems reusable across the project.

Gameplay Designer & Team Lead | Project Potion | Duration: 3 Months | Unreal Engine 5 (PaperZD) | Team Size: 7

- Designed and implemented all enemy AI behaviors, interactions, and attack patterns including a multi-phase boss encounter.
- Built a multi-phase boss (The General) with escalating movesets, a transition phase with minion waves, and phase-gated difficulty scaling.
- Engineered a custom projectile system shared across all potion types and enemy attacks via a common base class.
- Designed a damage system supporting modifiers, status effects, and sprite-flicker invulnerability-frame feedback.
- Implemented a full player movement suite (Dash, Wall Jump, Wall Slide) using PaperZD for 2D sprite animation support within Unreal Engine 5.
- Managed the Git repository and resolved merge conflicts across a 7-person team.
- Maintained Jira and Google Drive for task tracking and documentation; assigned roles and sprint tasks as Team Lead.

PROFESSIONAL EXPERIENCE

Cole Engineering, Inc. (A By Light Company) | Orlando, FL *Test Engineer 1* | Jun 2024 - Present

- **Automation Leadership:** Pioneered the test team's adoption of test automation by engineering a robust framework from scratch utilizing Python, Behave, BDD, and Gherkin.
- **Efficiency Gains:** Significantly reduced test execution time for critical test cases by **over 75%** (from 6m 26s manual execution to 1m 26s automated), integrating the solution with Bamboo and Cucumber for continuous integration.
- **DoD Compliance:** Designed and executed manual and automated test cases for a highly complex training tool under DoD contract, ensuring strict adherence to government specifications and high quality deliverables.
- **Data Analytics & Visibility:** Leveraged advanced JQL queries to develop custom live dashboards in Jira and Confluence, enhancing visibility into test progress and defect trends for stakeholders.
- **Performance Reporting:** Optimized insights on testing metrics by architecting specialized Power BI dashboards using DAX, ensuring precise adherence to specific government reporting standards. As well as setup notifies within CI/CD pipelines to send test reports to Xray and specific stakeholders when new builds are pushed out.

EDUCATION

Full Sail University | Orlando, FL *Bachelor of Science* | Aug 2022 – May 2024